

CoralScapes:
A Study into the Optimization of Coral Nursery Design

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ME 80: Mechanics of Materials
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1. Introduction

Coral reefs play a vital role in marine ecosystems. Since the 1950s, climate change, pollution, and anthropogenic activity have caused coral populations to decline by over 50% (Coral Restoration Foundation, 2025). Coral reefs have a \$9.9 trillion dollar estimated annual global economic value and they sustain the livelihoods of 3 billion people who depend on them for fishing, food, and eco-tourism (United Nations). Coral nurseries serve as intermediate growth areas where coral fragments are cultivated before transplantation to degraded reefs. In our project, we sought to optimize the structural design of a coral nursery to better withstand ocean currents, wave action, and environmental forces while optimizing conditions for coral growth and propagation processes. Our goals included: *1) optimizing structural stability to prevent damage from strong currents and storms; 2) Ensuring the ease of coral attachment and monitoring for researchers and conservationists; 3) Using environmentally friendly materials that minimize ecological impact while still maximizing durability.* This project and mission is highly relevant and important as coral restoration efforts rely on stable nurseries to grow healthy corals before transplantation can happen. If the nursery is unstable, it could fail structurally and lead to coral loss and wasted conservation efforts. With a gap in this research, we took this project on.

2. Design

The design process included thorough research into existing designs and what they best serve for. We would come to learn that coral nursery design and selection is partially dependent on location. We did research into four main types of coral nursery structures: the “Mars Star” Structure, Coral trees, Ropes Between Rebar, and Fixed Frame Structures. Initially, we hoped to base our research in the Bay Islands, specifically Roatan. We pivoted after learning that Bay Islands underwater landscape got deep very fast and was best suited for coral trees. We felt that a coral tress, a floating structure, was going to be too hard to test in a scaled-down approach. Thus, as intriguing as some ideas were that came up in initial design conversations, we decided to, instead, iterate on the classic, rectangular “bed,” fixed frame structure. Our first design choice was to make the frame hexagonal to distribute forces more easily. We also made the feet hexagonal rods, meant to help reduce drag coefficient and improve stability (buckling analysis

was out of scope), filleted edges, and included holes within the joint of the frame for the bed of the coral. For the bed, we wanted to implement a “chain-mail” like mesh system that is flexible rather than the rigid, squared, steel frames that are standard, to promote water and nutrient flow. The time and scope of this project did not lend itself enough time for a proper analysis or prototyping of this. However, as it was one of our original goals, we will briefly discuss research results that arised. The mesh would have been a system of interlocking holes. Within the hole slots, we would've liked to research three main techniques for attaching coral: 1) threaded easy-insert tubes; 2) Spiral easy-open-close mechanism; 3) Pressure held Bristle or Rubber technology (see illustrations below).

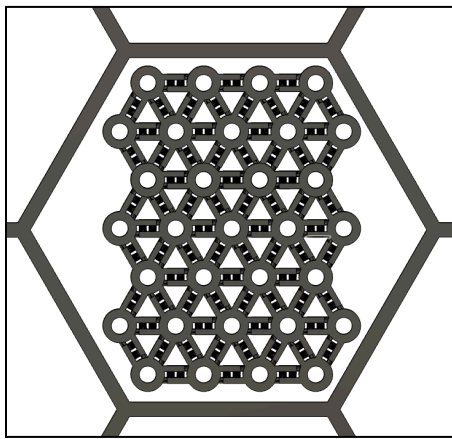


Figure 1: Mesh Prototype

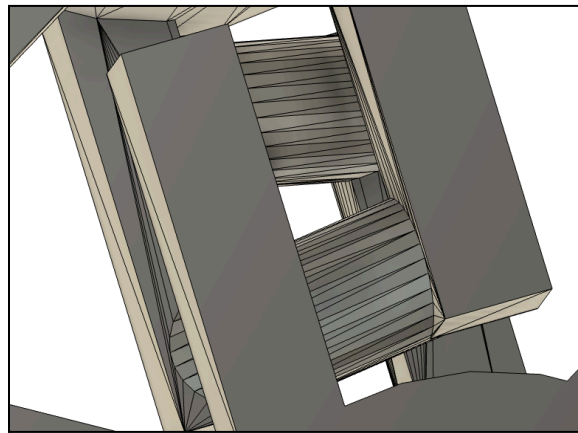


Figure 2: Close up of Interlocking design

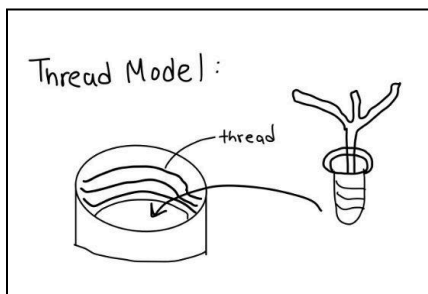


Figure 3: Thread Model Sketch

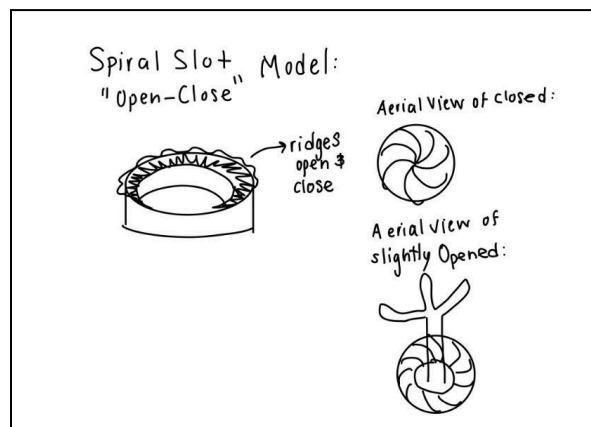


Figure 4: Spiral Slot Model Sketch

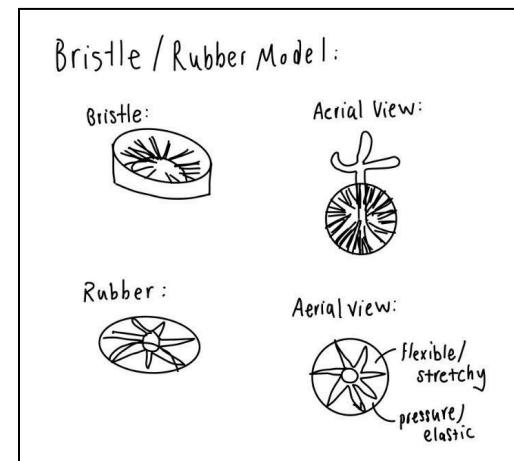


Figure 5: Bristle/Rubber Model Sketch

3. Analytical Results

Our CoralScape structure was originally designed to be 4 feet in diameter to be situated approximately 20 meters under water. To make this problem more approachable, our main assumptions included that we were dealing with uniform material properties. We also simplified and assumed that the hydrostatic pressure would be the primary vertical load. Drag force was modeled as a uniform horizontal load. Finally the assumption that the feet were rigidly fixed to the seabed. The FEA set up is pictured below and covered in Simulation Results (3). Our primary analysis criteria included beam bending, yield, ultimate failure, and factor of safety. We chose this criteria because they are essential to getting a full picture of the structural integrity and vitality of our CoralScape structure.. These decisions were made to best evaluate how the structure would interact with these forces in real life.

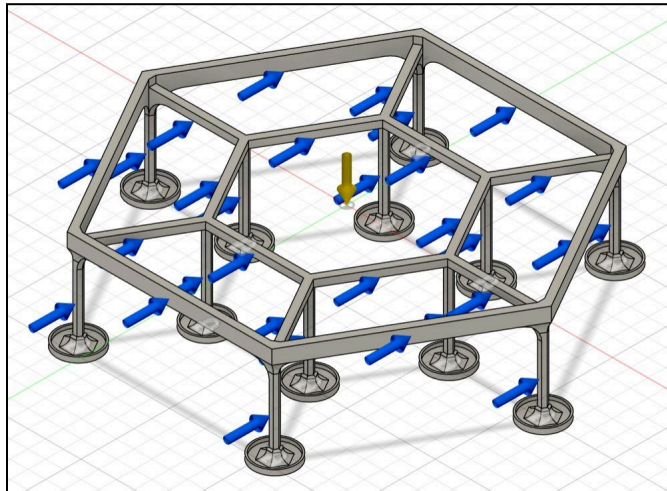


Figure 6: FEA Setup

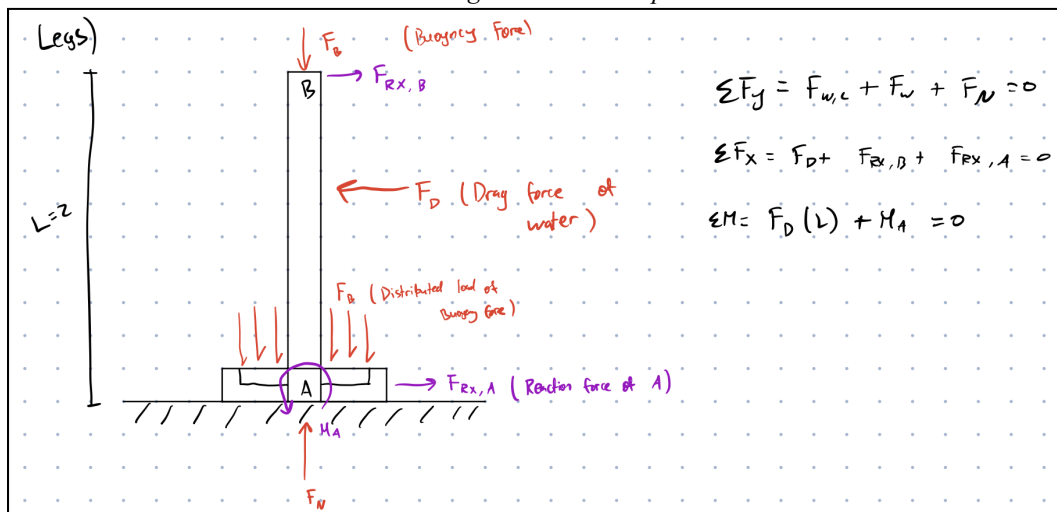


Figure 7: FBD of Legs

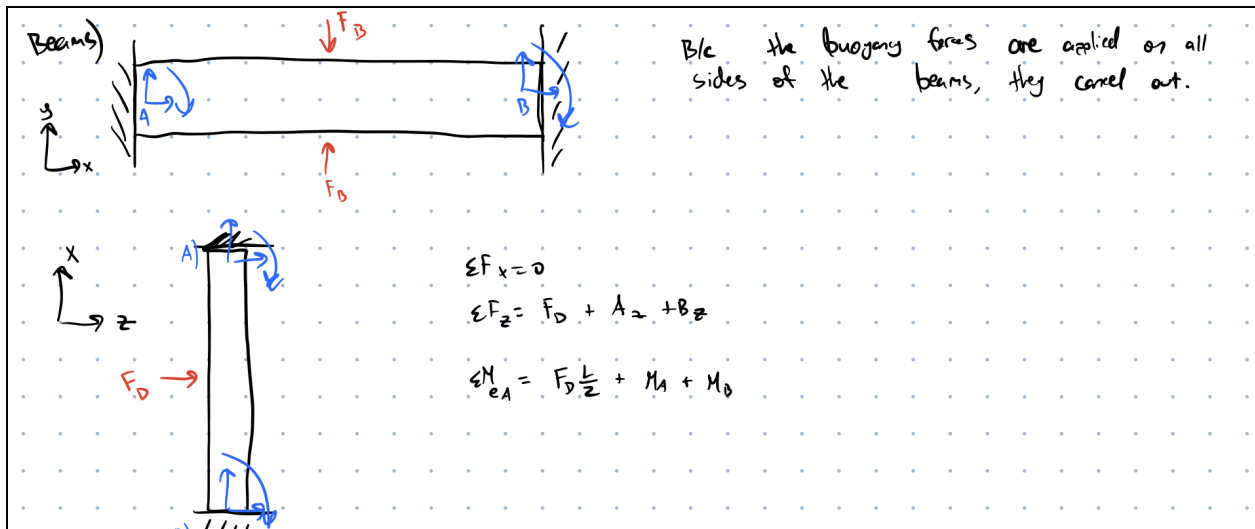


Figure 8: FBD of Beams

4. Simulation Results

We applied fixed constraints to our coral nursery at each of the 12 feet. We applied horizontal drag force to simulate water currents and buoyancy force vertically to simulate water pressure of being submerged 20 m below sea level (see appendix to see calculations). We wanted to model GFRP, so we selected this as our material on Fusion. Our FEA predicted high loads before yielding to brittle failure with the max stress concentration at the legs of the outer hexagonal frame. Similarly the lowest factor of safety of 1.45 was also found at the outer legs. However, modeling the individual beams involved in this design gave much more appealing results. The lowest factor of safety after testing the beams was 4.99.

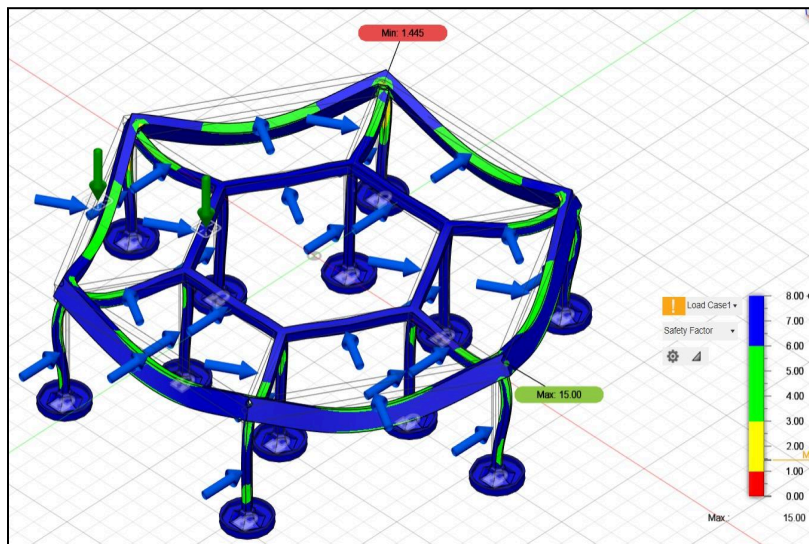


Figure 9: Full structure FEA

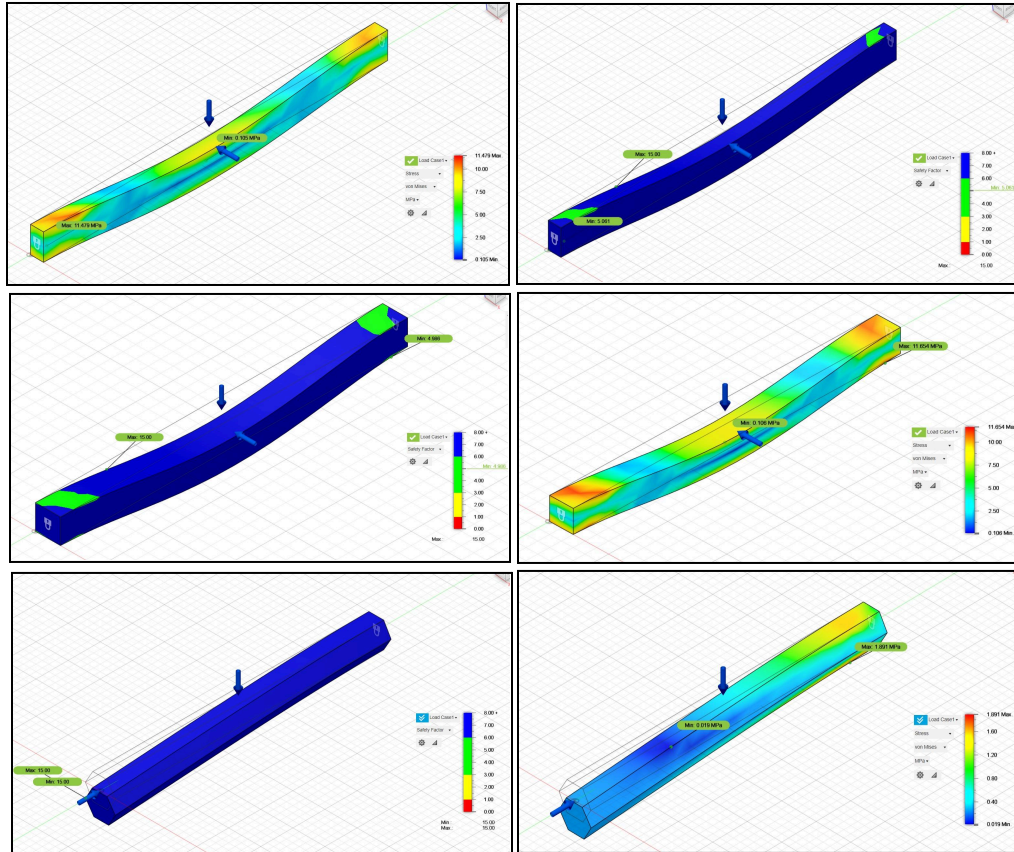


Figure 10: Individual feet FEA

5. Prototype and Testing

The prototype we tested on was designed to be 1/20 real life scale. The prototype we presented in our poster presentation was 1/15 the real life scale. Our real life results varied, we suspect, partially because of the anisotropic material of PLA. We conducted two sets of tests. The first test consisted of a straight downward force onto the full 1/20 hexagonal structure. Our procedure consisted of: 1) connecting three zip ties around a 9 inch diameter gear; 2) hooking a scale on the bottom zip tie; 3) separating two tables to have an empty wedge in between to pull from; 4) pulling and measuring.

Our second set of experiments were for the individual beams which were scaled to be $\frac{1}{10}$ the size. The experiment set up was the same as the first with two exceptions. The first being that the distance of the wedge in between the table did not remain the same. We noted this as an error in our experimental setup. The second was that we did not distribute the load and only applied it at a single point. We put a thick paper towel atop the beam to have the scale latch on easier to the beams (one test was done for the horizontal axis, the second test for the vertical test; results in appendix). Our testing did confirm that our prototype is weakest at the joints. Our FEA resulted in a max F.S. of 5 and min F.S. of 1.4 while our hand calculations resulted in max F.S. of 4.6 for the big beam and 2.2 for the small beam. When we set up and ran our FEA, our goal was trying to match the real life conditions as best as we could. However, our experimental goals differed in that we were trying to test the limits of the structure we had physically created. We decided to do this because we felt that the structure's real life conditions were more difficult to replicate via testing at this scale. However, in terms of our experiment, we are satisfied with the results and what they speak to the prototype ability at the force it can withstand.

6. Discussion

Our prototype testing confirmed that our design could withstand high loads even as PLA. GFRP is much more malleable so it would withstand more bending before it breaks. One larger experimental error we had was failing to keep the distance that we spread the tables apart constant for the horizontal and vertical axes testing constant or even measured. Because our prototype was only 3D printed the process of 3D printing was not that difficult. However, we did have to iterate on our design after finding how weak PLA can be if we don't properly scale or have thin joints.

Our FEA analysis gave appealing results despite the lowest factor of safety being just above 1. When designing our newer model, we weren't able to incorporate the supported joints that we created for our first design, likely resulting in the low factor of safety. Furthermore, we caught a small error in setting up the simulation. When applying the buoyancy force to simulate water pressure based on depth, we applied it as a distributed load across the entire top side of the model (green arrows in the picture). In reality, since the beams are completely surrounded by water, the pressure would cancel, so this pressure would only apply vertically above the legs.

Doing this in fusion would greatly reduce the amount of vertical force being applied to the model, which would increase the experimental factor of safety.

While analyzing the tests show that our design would most likely be suitable for withstanding the extremes of underwater forces, we weren't able to test for durability over time. The next step of tests would be to use fatigue testing and corrosion/environmental testing to see if our design would be able to withstand a long time underwater nurturing corals.

7. Final Product

Our research and testing process was a bit organic. Looking back, it might have been helpful to be more restrictive/careful on when and where decisions were made. We actually had a very first prototype that we have yet to talk about. We discovered fairly quickly that it was much too thin and that a printed joint system without the proper fiction attachment joints made the PLA structure too weak. The structure was broken after a half mile walk from AMPS to the dormitory. This drove for the change in regular rectangular beams as the feet to hexagonal feet. Also this made us add filleted frames into the feet atop the sand plates. After seeing this, we decided to pivot away from the joint idea (to facilitate an easier assembly process underwater) and to only focus on the structural integrity of the suture.

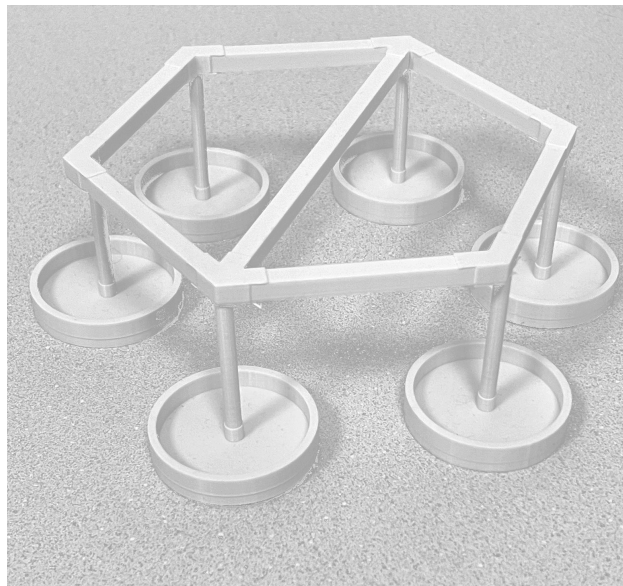


Figure 11: First Prototype

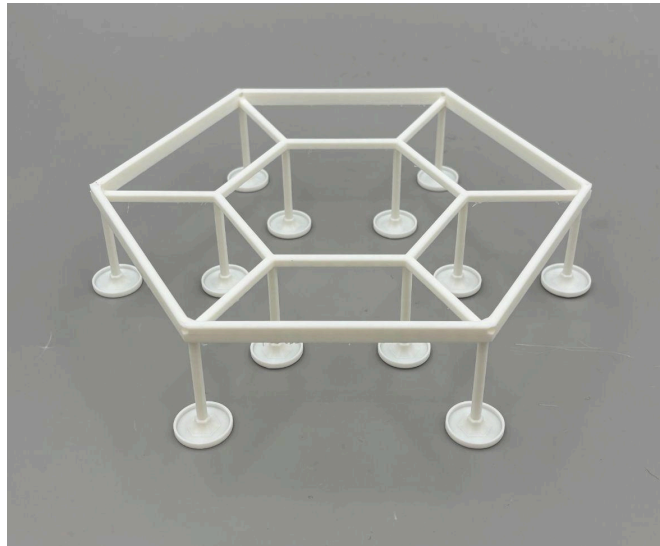


Figure 12: Final Prototype

8. Conclusion

One of the main pain points in our process was the back and forth on design. It felt that we were constantly discovering new things in our research process because we were constantly learning and researching throughout the entirety of the process. While this might have been beneficial in a long term study, for the short time frame of this project, this was not a good match. For future projects it is important to better dedicate a frame of research and solidify the design and limit, if at all, one redesign iteration (time-dependent). The material of GFRP was elected as an environmentally friendly alternative because we found that it requires less energy & emits less carbon during manufacturing than steel counterparts. We also have some key takeaways to implement in a future design iteration: 1) adding triangular supports in each joint that support each of the stemming three branches to increase the minimum factor safety; 2) not placing the mesh attachment holes to the joints but rather to the length of the beam where we have the highest safety factor; 3) Adding side hooks or “fins” to redirect water flow and minimize shear stress of the structure (something we would have liked to have tested). Also, of course, we would have loved to do further research into flexible chainmail-like mesh with slots for coral attachment. If we had more time and more training, we could have better accounted for shear stress and buckling. We might have also done a comparison between two types of structure instead of just one for a comparison analysis.

9. Peer Review

There were a lot of moving parts in our project. Roles become more fluid and intertwined to best achieve our results. In light of that, we broke up our peer review into phases of our project rather than roles.

Design and Research

Beals: Nico and Chloe jumped into this project early and learned a great deal about current coral nurseries, their strengths and weaknesses, and what factors we need to consider when designing our own. I'm glad they figured all this out early because when I had to design and model the thing, they made it very easy for me by just communicating what the best model would look like. Their work acted as the starting point and backbone of this project.

FEA

Nico: Beals did a really amazing job holding into account various real life conditions. He was super diligent and always was ahead of the curve. He made sure it was grossly overestimated to ensure a high factor of safety making the force equivalent to that of a tsunami.

Chlo: Michael did amazing at simulating complex conditions underwater and was able to come close to the stresses that our structure would experience. His understanding of the currents, etc. allowed him to optimize the shape of our structure to make it more durable. He was also really good at making any adjustments based on what structures for coral propagation have worked previously and which ones have not.

Hand Calcs

Beals: All I had to do was send Chlo the numbers and she did all the rest. Despite being on the other side of the country, she got all the necessary information done in a timely manner which made the whole process stress-free.

Nico: Chlo was amazing with the hand calculations. She cross checked her numbers alongside Beals to make sure the prototype would be the best it can be. Also a huge shoutout to Chloe for her knowledge on this subject area, it was super helpful and was immensely helpful as we made our way through it.

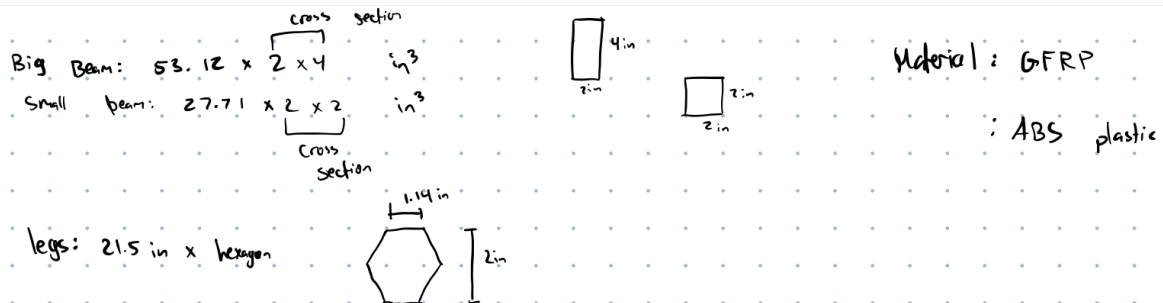
Simulation and Testing

Chlo: I was not able to be there for the testing of our structure but Michael and Nicole carefully set everything up and replicated the deep water conditions enough to make us confident in our design. It was smart of them to distribute the weight using gear. For future testing, they can keep track of the distance and test the pulling from the same location. I would've loved to be there but regardless it was so fun watching the videos of their results.

Overall, as a team we worked very well. We are all very passionate about the environment and nature and found ourselves finding ourselves in rabbit holes during our research process together. We all have our strengths in niche areas that serve the team functionality very well in this project.

Appendix

Finding Max Forces - Pre FEA Calculations



Big Beam:

$$* F_{\text{Drag}} = \frac{1}{2} C_D \rho A V^2$$

where: C_D for a rectangular block perpendicular to flow ≈ 2

$$\rho_{\text{seawater}} \approx 1035 \text{ kg/m}^3$$

$$A = h = 53.12 \times 4 = 212.48 \text{ in}^2 = 0.137 \text{ m}^2$$

$$V_{\text{avg}} \sim 0.6 \text{ m/s}$$

$$V_{\text{max (tunnel)}} \sim 2.4 \text{ m/s}$$

$$F_{D, \text{avg}} = 51.05 \text{ N}$$

$$F_{D, \text{max}} = 816.74 \text{ N}$$

} horizontal

$$P = \frac{F}{A} = \frac{816.74 \text{ N}}{0.137 \text{ m}^2} = 5961.61 \text{ Pa} = 5.96161 \text{ kPa}$$

$$* F_{\text{weight}} = \rho g h A \quad (\text{From ME 70})$$

$$\text{where: } \rho = 1035 \text{ kg/m}^3$$

$$g = 9.81 \text{ m/s}^2$$

$$h_{\text{max}} \approx 20 \text{ m}$$

$$A = 2 \times 53.12 = 106.24 \text{ in}^2 = 0.069 \text{ m}^2$$

$$\rho_{\text{air}} = 1.225 \text{ kg/m}^3$$

$$g = 9.81 \text{ m/s}^2$$

$$h$$

$$F_{w, \text{max}} = 14011.6 \text{ N} \Rightarrow P = F/A \Rightarrow P = (14011.6 \text{ N}) / 0.069 \text{ m}^2 = 203067 \text{ Pa} = 203.067 \text{ kPa}$$

Applied to Fusion: FEA: $F_{0.5} = 5.06$
 Max stress: 11.479 MPa



For $F_{D, \text{max}}$, $F_{w, \text{max}}$

Small Beam:

$$F_{\text{Drag}} = \frac{1}{2} C_D \rho A V^2$$

All same as above except for A

$$A_2 = 2 \times 27.21 \text{ in}^2 = 54.42 \text{ in}^2 = 0.0351 \text{ m}^2$$

$$F_{\text{Drag, avg}} = 13.08 \text{ N}$$

$$\star F_{\text{Drag, max}} = 209.252 \text{ N} \rightarrow P = F/A = 5961.6 \text{ Pa}$$

$$F_{\text{weight, max}} = \rho g h A \quad (\text{same as above, } A = A_2)$$

$$F_{w, \text{max}} = 7127.65 \text{ N} \Rightarrow P = F/A = 203067 \text{ Pa}$$

Applied to Fusion: FEA: $FoS = 4.99$
Max stress = 11.654 MPa

$\star$ Legs

$$F_{\text{Drag}} = \frac{1}{2} C_D \rho A V^2$$

C_D for a hexagonal cross section ≈ 1.3

$\rho_{\text{seawater}} \approx 1035 \text{ kg/m}^3$

bigger $\rightarrow A_1$ (side facing) = $2.309 \text{ in} \times 21.5 \text{ in} = 49.64 \text{ in}^2 = 0.032 \text{ m}^2$

A_2 (edge facing) = $2 \text{ in} \times 21.5 \text{ in} = 43 \text{ in}^2 = 0.0277 \text{ m}^2$

$V_{\text{avg}} \sim 0.6 \text{ m/s}$

bigger $\rightarrow V_{\text{max (tunnel)}} \sim 2.4 \text{ m/s}$

$$F_{D, \text{avg}} = 57.8 \text{ N}$$

$$\therefore F_{D, \text{max}} = 124 \text{ N} \Rightarrow P = F/A = 124 \text{ N} / 0.032 \text{ m}^2 = 3875 \text{ Pa}$$

$$F_{w, \text{max}} = \rho g h A, \quad (\text{same as above})$$

$$\text{where } A = \frac{3\sqrt{3}}{2} (1.155) = 3 \text{ in}^2 = 0.00194 \text{ m}^2$$

$$F_{w, \text{max}} = 393.95 \text{ N} \Rightarrow P = F/A = 203067 \text{ Pa}$$

Applied to fusion: FEA: $FoS = > 15$
Max stress = 1.891 MPa

Total stress:

$$P_w = 207067 \text{ Pa}$$



★ $F_0 \sin 30$ for slanted edges

FEA: Minimum FOS: 1.445

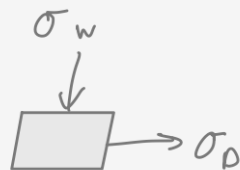
Max stress: 40.204 MPa

Key takeaways: FOS was around 5 at all areas except for the outer leg joints where we have a hole to tie the mesh to the frame. If we redesign both these features, we can increase the FOS to hopefully above 3.

Checking Results - Post FEA Calculations

Big Beam:

$$\sigma_y = 500 \text{ MPa}$$



$$F_{Wmax} = 14011.6$$

$$53.12 \text{ inches} \rightarrow 1.3492 \text{ m}$$

$$2 \text{ in} \rightarrow 0.0508 \text{ m}$$

$$\begin{aligned} \sigma_{bend} &= \frac{-Mc}{I} = \frac{-14011.6 \text{ N} \left(\frac{1.3492 \text{ m}}{2} \right) (0.0508)}{\frac{(0.0508)(0.1016)^3}{12}} \\ &= -108.15 \text{ MPa} \end{aligned}$$

$$F.S = \frac{500}{108.15} = 4.623$$

Small Beams:  2 in $L = 27.71$ in
2 in

Small beam

$$\sigma_{\text{bend}} = \frac{-14011.6 \text{ N} \left(\frac{0.703 \text{ m}}{2} \right) (0.0254)}{\frac{(0.0508)(0.0508)^3}{12}} = -225.4 \text{ MPa}$$

$$F.S. = \frac{500}{225.4} = 2.21$$

References

Coral Restoration Foundation. (2025). *Why save coral reefs?*

<https://coralrestoration.org/why-save-coral-reefs/>

United Nations. (n.d.). *Oceans and seas*. Sustainable Development Goals. Retrieved March 15, 2025, from <https://sdgs.un.org/topics/oceans-and-seas>